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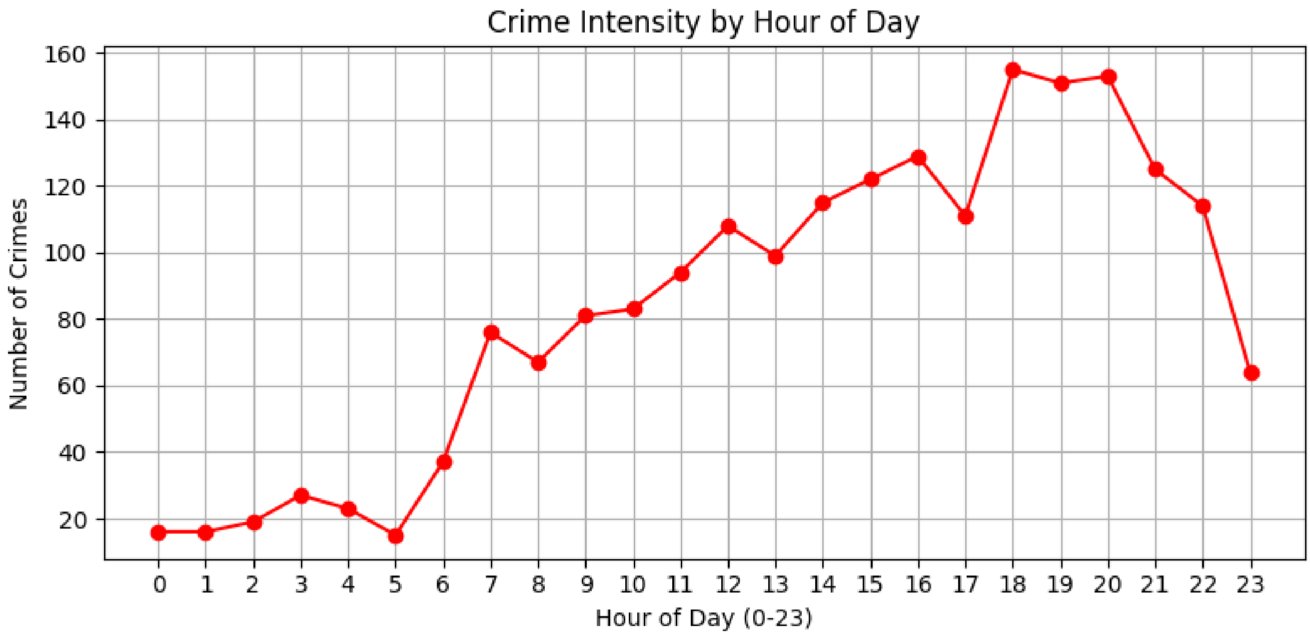
Use Case 3 — Statistical Insights & Pattern Detection

Objective: Use numerical and visual analysis to derive deeper insights.

Tasks

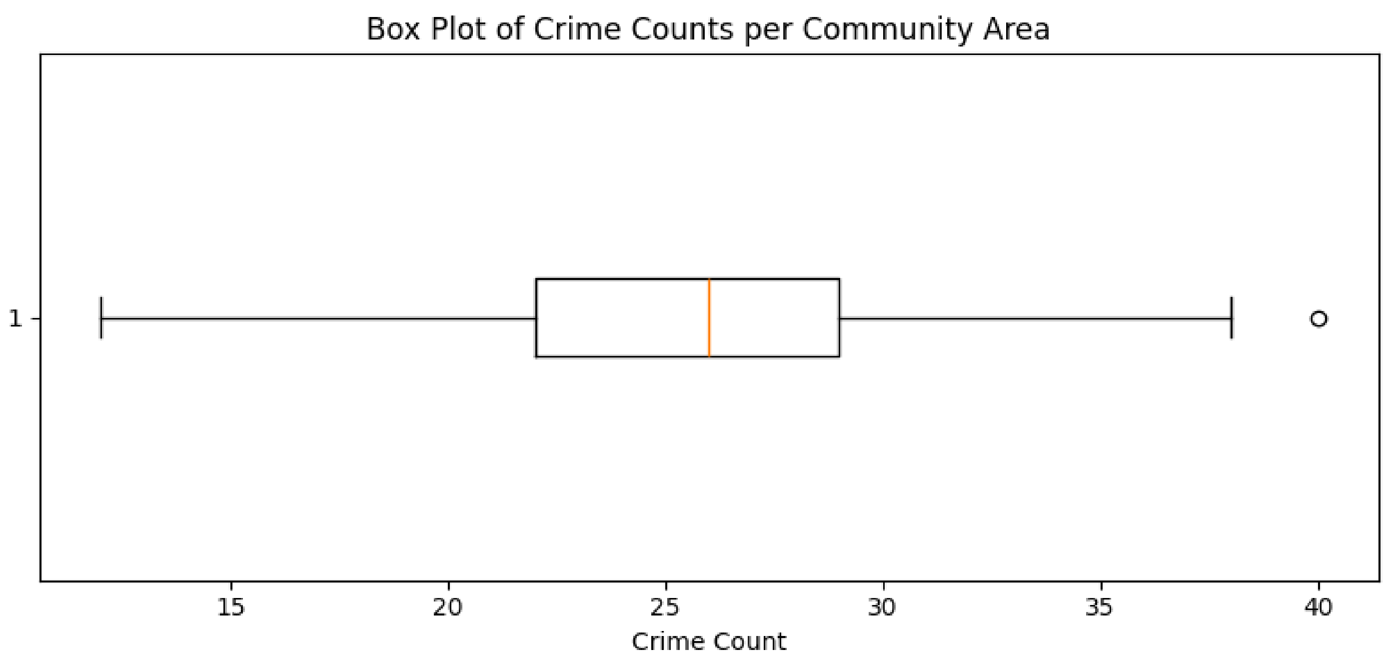
1. Crime Intensity by Time

- Use Pandas groupby to get hourly distribution.
- df['Hour'] = df['Date'].dt.hour
- crimes_by_hour = df.groupby('Hour').size()
- Line plot of crimes per hour of day.



2. Community Area Clusters Using NumPy

- Use NumPy functions to compute mean crime per community area.
- Identify which areas are extreme outliers in terms of crime counts(Use box plot to identify the outliers and then IQR method to list them)



```
Mean crimes per community area (NumPy): 25.51
Q1: 22.0, Q3: 29.0, IQR: 7.0, Upper Bound: 39.5
Extreme Outlier Communities (IQR Method):
community_code
56    40
Name: count, dtype: int64
```

3. Crime Cross-Correlation

- Use Pandas .corr() on numeric features (Year, Month, Arrest, etc.)
- Visualize correlation matrix with a heatmap.